Lab Manual: Practical 7

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Create a Basic AR or VR Mini-Experience Using Unity and XR Interaction Toolkit

Duration: 4 Hours

Lab Type: Immersive Technology Integration

Assessment Weightage: Part of CAL 1 (Lab Assignments)

Objective

To design and implement a basic Augmented Reality (AR) or Virtual Reality (VR) experience using Unity and the XR Interaction Toolkit. Students will learn how to build immersive environments and interactions suitable for AR/VR platforms.

Pre-requisites

- Familiarity with Unity Editor and GameObject manipulation
- Basic C# scripting knowledge
- Understanding of AR/VR concepts and device capabilities
- Exposure to Unity's XR Interaction Toolkit and input systems

Learning Outcomes

By the end of this lab, students will be able to:

- Set up Unity for AR or VR development using XR plugins
- Create interactive scenes for AR object placement or VR exploration
- Implement basic user interactions like viewing objects
- Deploy and test the experience on supported devices or simulators

Tools & Resources

- Unity Editor (2021.3 LTS or later recommended)
- XR Interaction Toolkit (via Unity Package Manager)
- AR Foundation (for AR projects)
- VR headset (e.g., Oculus Quest) or AR-capable mobile device (optional)
- Reference: Unity Learn – XR Development

Lab Tasks

Task 1: Project Setup

- Create a new Unity project (3D template)
- Install XR Interaction Toolkit and configure input settings
- Choose AR mode and set up required packages (AR Foundation)

Task 2: Scene Design

- For AR:
    - Add AR Session and AR Session Origin
    - Implement object placement using touch input
    - Use plane detection and anchors

Task 3: Interaction Implementation

- Use XR components.
- Write C# scripts for custom interactions (e.g., object scaling, color change on touch)
- Add UI elements for feedback or instructions

Task 4: Testing & Deployment

- Test the experience in Unity Editor or on device
- Debug interaction issues and optimize performance
- Document the experience with screenshots or video

Evaluation Criteria

Component	Marks
XR Setup & Configuration	3
Scene Design & Immersion	3
Interaction Implementation	2
Code Quality & Testing	2
Total	**10**

Tips for Success

- Use Unity's sample scenes to understand XR workflows
- Keep the experience simple and intuitive for first-time users
- Test frequently to ensure compatibility with devices
- Use lighting and sound to enhance immersion

Suggested Reading & Resources

- *Unity Game Development Essentials* by Will Goldstone
- Unity Learn: Getting Started with XR
- YouTube: Unity XR Toolkit tutorials by Valem or Dilmer Valecillos

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.
- Submit a video.